

2) a) Given: required load to crack nut: 222.18 kg or 2198 N

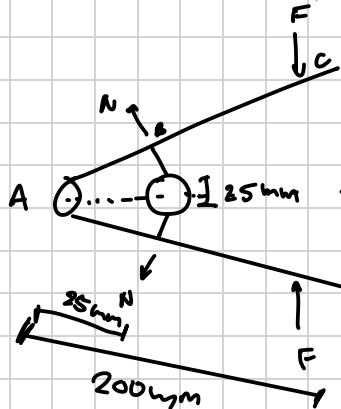
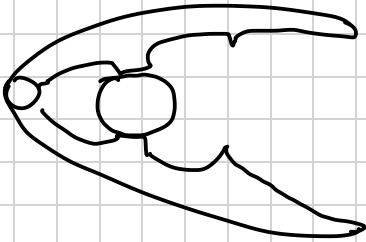
Grip strength: 300 N

Size of nut: 25mm

Find: Dimensions of nutcracker

Plan: FBD, moment balance, determine L ratio, determine H ratio

Solution:



$$F_{\text{grip}} = 300 \text{ N}$$

$$N = 222.18 \text{ kg} \\ = 2198 \text{ N}$$

$$MA = \frac{2198}{300} = 7.26 \rightarrow 8$$

$$L_j = \frac{L_n}{MA} = \frac{200 \text{ mm}}{8} = 25 \text{ mm}$$

$$L_L = 200 \text{ mm}$$

$$L_j = 25 \text{ mm}$$

b) If $MA=8$, you need a force of $\sim 500 \text{ N}$, which is the average grip strength. Could make handle longer to make it easier

$$c) F_{24} = \frac{2198}{8} \approx 272 \text{ N} \sim \text{or } 100 \text{ lbf}$$

Need actuator $\sim 2 \text{ in}$ strokes, 100 lb

1965, stroke: 1-4 in, force: 16-225 lb

$$\sum M = L_L \times F - L_N \times F_N$$

$$\frac{L_L}{L_N} = \frac{F_N}{F} = \frac{222.18 \text{ kg}}{225 \text{ lbs} \cdot \frac{1 \text{ kg}}{2.2 \text{ lbs}}}$$